

App Development Lab Report: Thumblytics

1. Project Overview

- **Project Title:** Increase your Clicks with AI using **Thumblytics**
- **App Concept:** A specialized thumbnail analyzer designed to audit and optimize images for social media platforms.
- **Purpose:** To address low Click-Through Rates (CTR) caused by suboptimal thumbnail design. Thumblytics identifies design flaws, provides comprehensive performance reports, and suggests improvements to help content creators increase views and impressions.

2. Technical Methodology

- **AI Tools Used:** Google AI Studio
- **Development Approach:**
 - **Prompt Engineering:** Ideas were initially refined through a secondary AI model to generate high-fidelity technical prompts.
 - **Iterative Development:** The project utilized an iterative "Debug-and-Patch" cycle, using error logs and feature-specific prompts to guide the AI.

3. Prompt Iteration & Development

The development process relied on precise, issue-driven prompts:

Feature/Issue	Prompt Strategy
Authentication	<i>"When I sign up to the app, it shows a 'failed to fetch data' error. Fix it."</i> (Resulted in Supabase key integration).
Data Export	<i>"The report button is not downloading data. Make it functional."</i> followed by <i>"Add options to export reports in .docx, .pdf, and .xlsx formats."</i>

4. Performance Analysis

- **What Worked Well:** Leveraging a two-step AI process—where one model designs the prompt and Google AI Studio executes it—resulted in a 95% functional build in a single attempt.
- **What Did Not Work:** The initial development phase was smooth; however, the primary hurdle was not the AI generation, but the external configuration of backend services.

5. Challenges & Resolutions

- **The Challenge:** The most significant difficulty was the configuration of **Supabase**. Integrating API keys and correctly mapping the backend to the frontend within the AI studio environment proved complex for the development workflow.
- **The Resolution:** Utilizing the AI as a pedagogical tool. By asking the AI to walk through the creation of Supabase keys, URL mapping, and database integration step-by-step, the development process evolved into a learning experience that bridged the gap between code generation and backend architecture.

6. Project Outcomes

- **Key Features Implemented:**
 - **Performance Metrics:** Average Click Potential and Best Score algorithms.
 - **Analytical Tools:** A/B testing comparison.
 - **Export Functionality:** Multi-format reporting (PDF, Excel, Docx).
- **Final Result:** A fully operational AI tool that analyzes thumbnail images, identifies strengths/weaknesses, and provides actionable data to improve creator reach.
- **Real-World Impact:** By providing creators with data-driven feedback, Thumblytics directly addresses the challenge of stagnant video performance by optimizing CTR and visual engagement.